



Detailed Lecture Notes: Probability vs. Likelihood

Video: [In Statistics, Probability is not Likelihood.](#)

Channel: StatQuest with Josh Starmer

1. Probability (Area under the curve)

Probability deals with predicting the **data** given a **fixed distribution**.

A. Definition and Focus

- **Probability** is the **area under the curve** of a fixed distribution [\[04:00\]](#).
- It answers the question: "What is the chance of observing data within a specific range?"
- It is used when the characteristics of the population (the distribution) are **known and fixed**.

B. Example: Mouse Weights

Imagine a **Normal Distribution** of mouse weights with the following fixed parameters:

- Mean (μ): 32 grams [\[00:54\]](#)
- Standard Deviation (σ): 2.5 grams [\[00:58\]](#)

The **Probability** is the area you are looking for:

- **Question:** What is the probability that a randomly selected mouse will weigh **between 32 and 34 grams**? [\[01:09\]](#)
- **Answer:** The probability is the **area under the curve** between 32g and 34g.
- **Result:** In this case, the area equals 0.29, or a **29% chance** [\[01:26\]](#).

C. Mathematical Notation

The mathematical notation for probability is written as:

$$P(\text{Data} \mid \text{Distribution})$$

- $P(\text{Weight between 32 and 34 grams} \mid \mu = 32, \sigma = 2.5) = 0.29$ [\[01:36\]](#)

Key Idea of Probability: The distribution parameters (μ and σ) on the right side of the equation are **fixed**, and the data on the left side is what you change to calculate a new area/probability [\[02:13\]](#).

2. Likelihood (Y-axis value for fixed data)

Likelihood deals with estimating the **distribution** given a **fixed data point**.

A. Definition and Focus

- **Likelihood** is the **Y-axis value** (the height) of the curve for a fixed data point [\[04:18\]](#).
- It answers the question: "How well does a specific distribution explain the single observed data point?"
- It is used when the measurement (the data) is **known and fixed**, and the distribution's parameters are what you change/test.

B. Example: Mouse Weights

Assume you have already weighed a mouse, and this measurement is fixed:

- **Data (Measurement):** 34 grams [03:01]

The **Likelihood** is the height of the curve at that data point:

- **Scenario 1:** Testing a distribution with $\mu = 32$ and $\sigma = 2.5$.
 - **Likelihood:** The height of the curve at 34g is 0.12 [03:11].
- **Scenario 2:** Testing a distribution with a shifted mean ($\mu = 34$) and $\sigma = 2.5$.
 - **Likelihood:** The height of the curve at 34g is now 0.21 [03:36].

C. Mathematical Notation

The mathematical notation for likelihood is written as:

$$L(\text{Distribution} \mid \text{Data})$$

- $L(\mu = 32, \sigma = 2.5 \mid \text{Weight} = 34 \text{ grams}) = 0.12$ [03:17]

Key Idea of Likelihood: The data on the right side of the equation is **fixed**, and the distribution parameters (μ and σ) on the left side are what you modify to find the best fit (Maximum Likelihood) [03:47].

3. Summary Comparison

Concept	What is Fixed?	What is Calculated?	Mathematical Notation	Interpretation [04:00]
Probability	The Distribution (e.g., μ, σ)	The Area under the curve (a chance)	$P(\text{Data} \mid \text{Distribution})$	Probability of getting a range of data points <i>given</i> a distribution.
Likelihood	The Data (the measurement)	The Y-axis Value (the height)	$L(\text{Distribution} \mid \text{Data})$	Likelihood of a distribution <i>given</i> the fixed data point.

http://googleusercontent.com/youtube_content/3